

Diabetes Prediction Using Machine Learning

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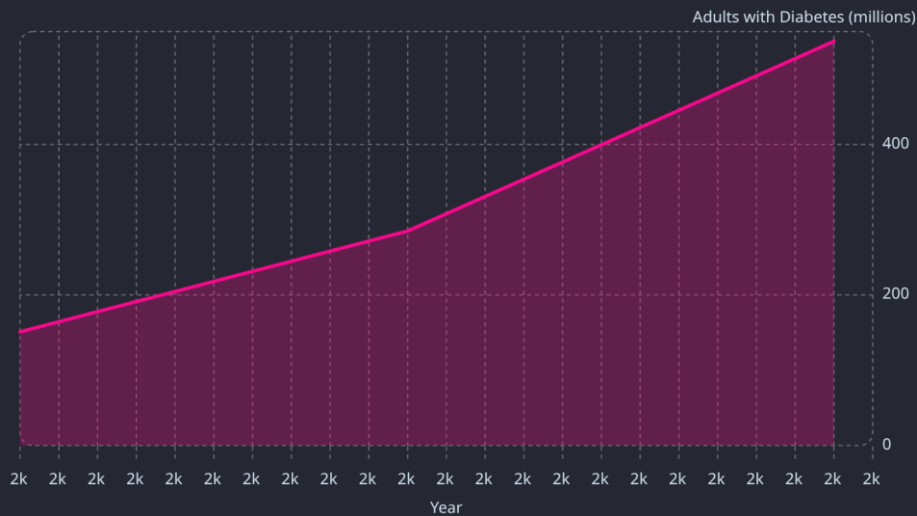
Early Detection of Diabetes with Artificial Intelligence



Introduction

Diabetes is one of the most common chronic diseases worldwide, affecting millions of people every year. Early detection plays a vital role in preventing serious complications and improving patients' quality of life.

Machine Learning provides intelligent techniques that analyze medical data and help predict diabetes risk quickly and accurately, supporting healthcare professionals in making informed decisions.



Global diabetes cases have more than quadrupled since 2000 — AI-powered tools can flag risk **before symptoms appear**.





Problem Statement & Objectives

1

Reactive Diagnosis Problem

Traditional diagnosis waits for late-stage symptoms to appear.

2

Build a High-Accuracy Model

Compare multiple ML algorithms head-to-head on clinical data.

3

Clinical Decision Support

Assist clinicians with data-driven, evidence-based risk assessment.

4

Earlier, Faster, Cheaper

Enable proactive diagnosis before costly complications arise.

Pima Indians Diabetes Dataset



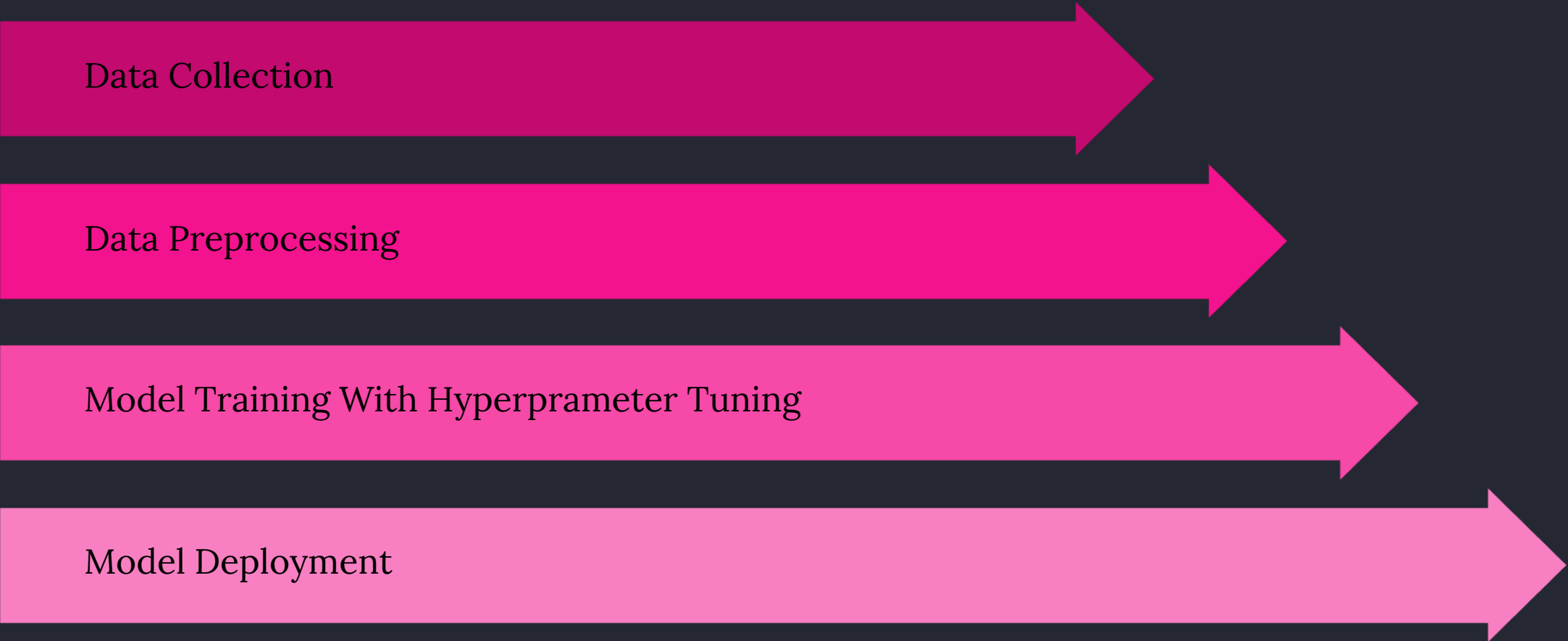
768 patient records · 8 clinical input features

Source: National Institute of Diabetes and Digestive and Kidney Diseases

- Glucose · BMI · Age · Insulin
- Blood Pressure · Pregnancies · Skin Thickness · Diabetes Pedigree Function
- Target: **Outcome** — Diabetic (1) or Non-Diabetic (0)

Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	DiabetesPedigree Function	Age	Outcome
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1

Methodology Pipeline



The diagram illustrates a four-stage methodology pipeline. Each stage is represented by a horizontal arrow pointing to the right, with the stage name written inside. The arrows are stacked vertically and increase in length from top to bottom. The colors of the arrows are a dark magenta, a bright magenta, a medium magenta, and a light pink, respectively.

Data Collection

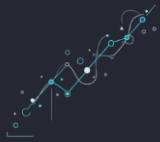
Data Preprocessing

Model Training With Hyperparameter Tuning

Model Deployment

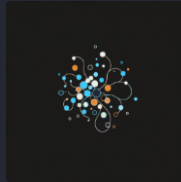
Built entirely with **Python** using Pandas · NumPy · Matplotlib · seaborn · Scikit-learn · Optuna · Streamlit · Joblib Each stage is reproducible and modular for clinical integration.

Machine Learning Models Compared



Logistic Regression

Linear baseline classifier for binary outcomes.



K-Nearest Neighbors

Proximity-based classification among similar cases.



Support Vector Machine

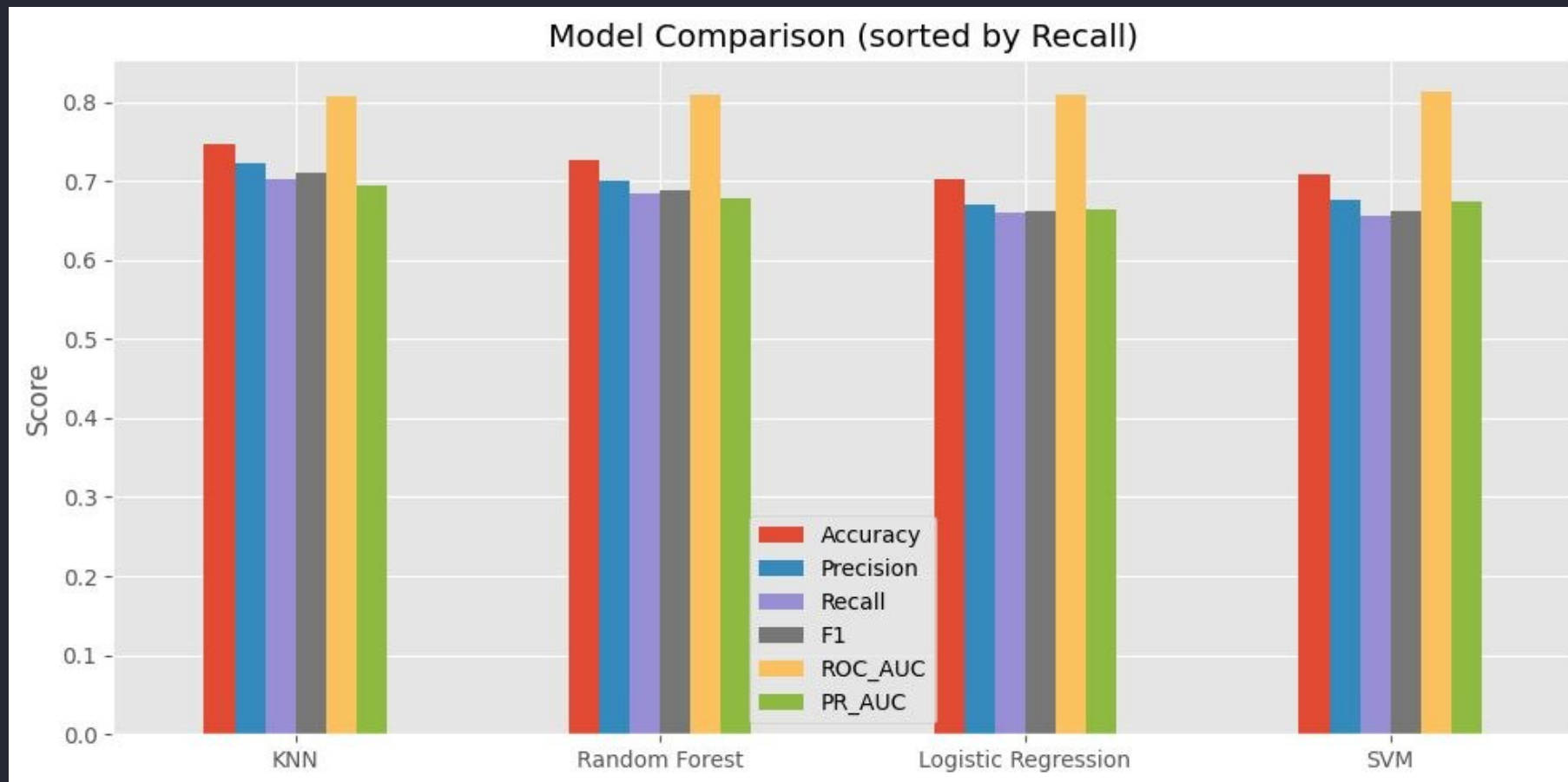
Optimal hyperplane separation for maximum margin.



Random Forest

An ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting

Model Evaluation Results



These metrics provide a comprehensive assessment of each model's prediction performance and reliability.

0.81

ROC-AUC Score
Excellent discrimination ability

GUI

A user-friendly graphical interface was developed to make the prediction process simple and interactive.

The application allows users to:

- Enter patient medical information.
- Predict diabetes risk instantly.
- Display the prediction result in a clear and easy-to-understand format.

Glucose, BMI, and Age are the top three predictive features driving model decisions.

Diabetes Risk Predictor

Choose a model

Logistic Regression

Pregnancies: 1

Glucose: 120.00

BloodPressure: 70.00

SkinThickness: 20.00

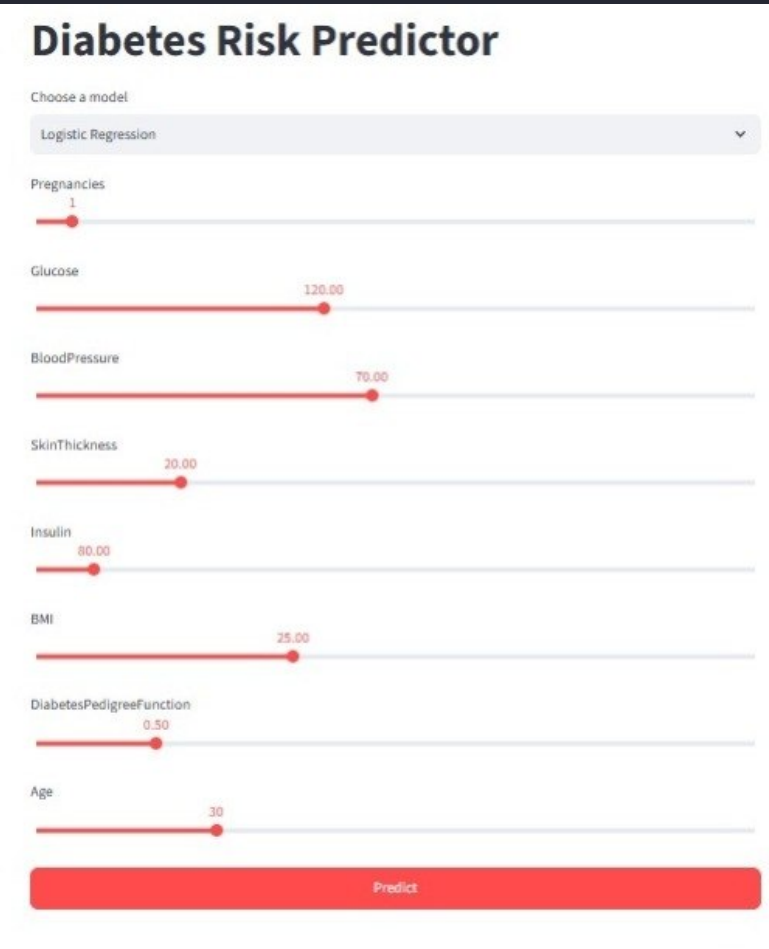
Insulin: 80.00

BMI: 25.00

DiabetesPedigreeFunction: 0.50

Age: 30

Predict

The image shows a web-based interface for a Diabetes Risk Predictor. It features a title 'Diabetes Risk Predictor' at the top. Below the title is a dropdown menu labeled 'Choose a model' with 'Logistic Regression' selected. The interface contains eight horizontal sliders for different medical features: 'Pregnancies' (set to 1), 'Glucose' (set to 120.00), 'BloodPressure' (set to 70.00), 'SkinThickness' (set to 20.00), 'Insulin' (set to 80.00), 'BMI' (set to 25.00), 'DiabetesPedigreeFunction' (set to 0.50), and 'Age' (set to 30). Each slider has a red dot indicating the current value. At the bottom of the form is a large red button labeled 'Predict'.

Applications & Future Directions

Now — Deployed Today

Hospitals & Clinics

Real-time clinical decision support at point of care.

Mobile Health Apps

Patient-facing risk screening on smartphones.

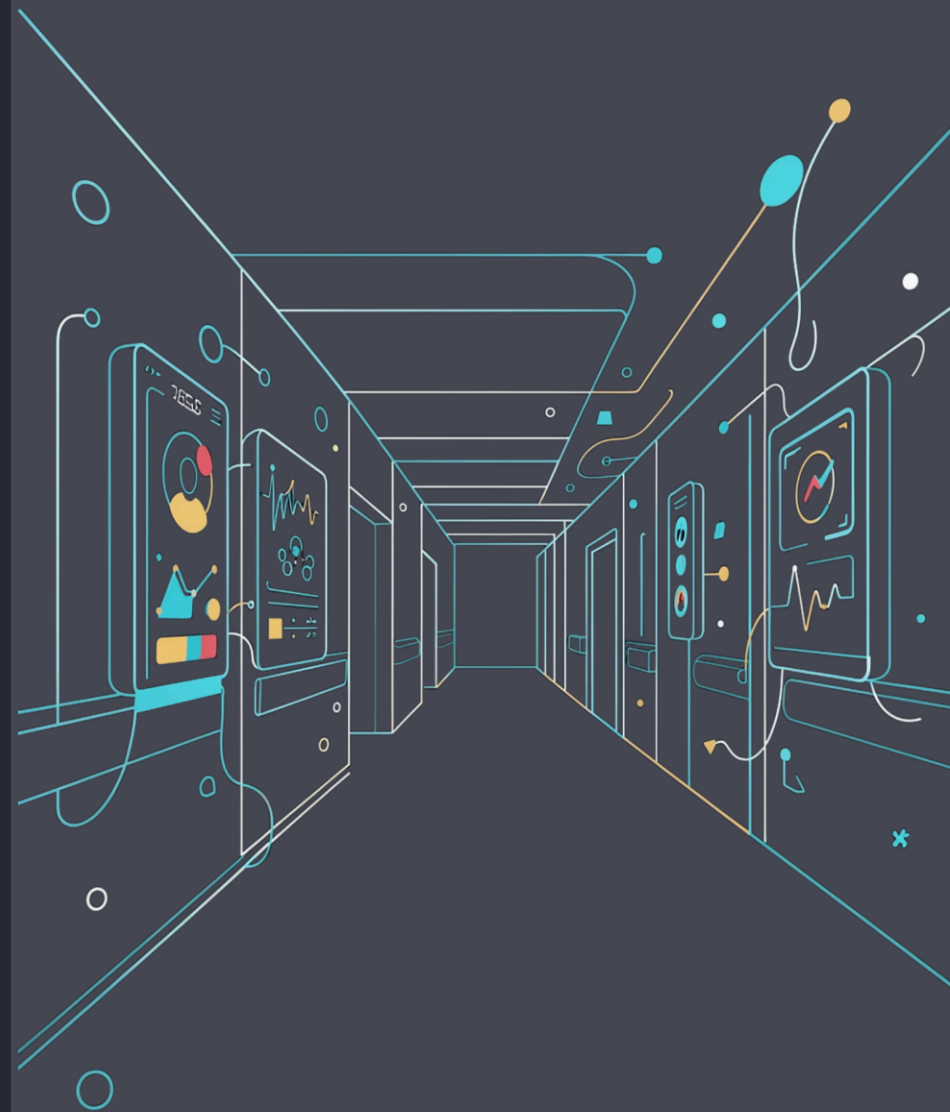
Next — On the Horizon

Deep Learning

Neural networks for higher predictive accuracy.

Explainable AI (SHAP)

Clinical transparency — knowing *why* a prediction is made.





Conclusion

This project demonstrates how Machine Learning can be used to support the early prediction of diabetes risk. By comparing multiple classification algorithms and developing a simple prediction interface, the system provides an efficient and practical tool that can assist healthcare professionals in making faster and more informed decisions.

Thank You